

Numpy

Aquest notebook és una versió adaptada d'Abhay Parashar (<https://www.kaggle.com/code/abhayparashar31/best-numpy-functions-for-data-science-50>).

===== TAULA DE CONTINGUTS =====

- Operacions de Conjunts
- Apilament
- Comparació de Dos Arrays
- Anàlisi Estadística

```
In [1]: import numpy as np
```

Apilament

Apilament Horitzontal

`hstack` apila (afegeix) un array al final d'un altre.

```
In [6]: a = np.array([1, 2, 3, 4, 5])  
        b = np.array([1, 4, 9, 16, 25])  
        np.hstack((a, b))
```

```
Out[6]: array([ 1,  2,  3,  4,  5,  1,  4,  9, 16, 25])
```

Apilament Vertical

`vstack` apila un array damunt d'un altre.

```
In [7]: a = np.array([1, 2, 3, 4, 5])  
b = np.array([1, 4, 9, 16, 25])  
np.vstack((a, b))
```

```
Out[7]: array([[ 1,  2,  3,  4,  5],  
               [ 1,  4,  9, 16, 25]])
```

Anàlisi Estadística

Mitjana

Calcula la mitjana de l'array.

```
np.mean(a, axis=None, dtype=None, out=None)
```

In [8]:

```
a
```

Out[8]:

```
array([1, 2, 3, 4, 5])
```

In [9]:

```
np.mean(a)
```

Out[9]:

```
np.float64(3.0)
```

```
In [ ]: a = np.array([[1, 2, 3], [5, 8, 4]])  
np.mean(a, axis=0)
```

```
Out[ ]: array([3. , 5. , 3.5])
```

```
In [ ]: np.mean(a, axis=1)
```

```
Out[ ]: array([2.          , 5.66666667])
```

Mediana

Retorna el valor mediana de l'array.

```
np.median(a, axis=None, out=None)
```

```
In [ ]: a = np.array([[1, 2, 3], [5, 8, 4]])  
np.median(a)
```

```
Out[ ]: np.float64(3.5)
```

Desviació estàndard i variància

```
In [ ]: a = np.array([[1, 2, 3], [5, 8, 4]])
```

```
In [16]: np.std(a)
```

```
Out[16]: np.float64(2.266911751455907)
```

```
In [17]: np.var(a)
```

```
Out[17]: np.float64(5.138888888888888)
```

Nombres Especiales

```
In [18]: # Null, NaN, desconegut  
np.nan
```

```
Out[18]: nan
```

```
In [19]: a = np.array([3, 2, 6, 1], dtype=float)  
a[2] = np.nan  
a
```

```
Out[19]: array([ 3.,  2., nan,  1.])
```



```
In [20]: # Infinit  
np.inf
```

```
Out[20]: inf
```

```
In [21]: np.inf > 1000000000000000
```

```
Out[21]: True
```

```
In [22]: # -Infinit  
-np.inf
```

```
Out[22]: -inf
```

```
In [23]: -np.inf < 0.0000000000000001
```

```
Out[23]: True
```

Datetime i Timedelta

```
In [24]: day = np.datetime64('2023-09-01')  
day
```

```
Out[24]: np.datetime64('2023-09-01')
```

```
In [25]: date_time = np.datetime64('2023-09-01 13:00')  
date_time
```

```
Out[25]: np.datetime64('2023-09-01T13:00')
```

```
In [26]: # Afegir 3 hores i 15 minuts  
date_time + np.timedelta64(3, 'h') + np.timedelta64(15, 'm')
```

```
Out[26]: np.datetime64('2023-09-01T16:15')
```

```
In [27]: # Crear un array de dates  
np.array(['2007-07-13', '2006-01-13', '2010-08-13'],  
         dtype='datetime64[D]')
```

```
Out[27]: array(['2007-07-13', '2006-01-13', '2010-08-13'], dtype='datetime64[D]')
```

```
In [28]: # Rang de datetimes  
np.arange('2023-09-01 13:00',  
          '2023-09-02 13:00',  
          dtype='datetime64[h]')
```

```
Out[28]: array(['2023-09-01T13', '2023-09-01T14', '2023-09-01T15', '2023-09-01T16',  
                '2023-09-01T17', '2023-09-01T18', '2023-09-01T19', '2023-09-01T20',  
                '2023-09-01T21', '2023-09-01T22', '2023-09-01T23', '2023-09-02T00',  
                '2023-09-02T01', '2023-09-02T02', '2023-09-02T03', '2023-09-02T04',  
                '2023-09-02T05', '2023-09-02T06', '2023-09-02T07', '2023-09-02T08',  
                '2023-09-02T09', '2023-09-02T10', '2023-09-02T11', '2023-09-02T12'],  
          dtype='datetime64[h]')
```